

Assignment 4

Deadline: 9th of August, 2025 at 11:59. All team members submit on teams.

Problem 1: Conceptual Questions

Answer the following questions with references and at least one figure per each question or one diagram.

◆ Theoretical Assignment Questions

- ◆ 1. Why is MQTT often preferred over HTTP in IoT applications, especially with constrained devices like ESP32?
- ◆ 2. Compare MQTT and WebSocket in terms of message delivery style and typical use cases in IoT systems. When would each be ideal?
- ◆ *3. *What are the trade-offs between using HTTP vs MQTT for an IoT weather station? Include latency, reliability, and architecture needs.*
- ◆ 4. In what IoT scenario would WebSocket be a better option than MQTT or HTTP, and why?
- ◆ 5. Explain the role of a "broker" in MQTT. Why is it necessary, and how does it differ from client-server HTTP communication?
- ◆ 6. What is the difference between short-range communication protocols (like Wi-Fi and Bluetooth) and long-range protocols (like LoRa and NB-IoT) in terms of power, range, and bandwidth?
- ◆ 7. Why would you choose LoRa over Wi-Fi for a farm-wide soil moisture monitoring IoT system?
- ◆ 8. Compare LAN (Ethernet) with Wi-Fi for IoT devices inside a factory. What are the pros and cons for each?
- ◆ 9. What advantages does NB-IoT offer over traditional cellular (3G/4G) or Wi-Fi in smart city applications?

♦ *10. For an IoT-based health tracker worn by a patient, which communication technology would you use: Bluetooth, Wi-Fi, or LoRa? Justify your choice in terms of latency, power, and distance.

✓ Task 1: ESP32 + NTP Server

🧠 Objective:

Use ESP32 to fetch real-time data from an NTP server and display it.

🔧 Tasks:

- Connect ESP32 to Wi-Fi.
- Fetch time using an NTP server.
- Display time on Serial Monitor (LCD optional).
- Handle connection failures and retry logic.

📦 Deliverables:

- Working code
- Brief report (how NTP works, steps, issues)
- Max 3-min demo video
- Circuit design
- Wokwi simulation

✓ Task 2: ESP32 Web Server for LED Control

🧠 Objective:

Control two LEDs via a simple web interface hosted by the ESP32.

🔧 Tasks:

- Connect ESP32 to local Wi-Fi.
- Host a basic web page (HTML + CSS).
- Control LEDs on GPIO 26 & 27 using buttons.

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Deliverables:

- Working code
 - Brief report
 - Max 3-min demo video
 - Circuit design
 - Wokwi simulation
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Task 3: ESP32 Sensor to HiveMQ MQTT

Objective:

Read a sensor value and publish it to HiveMQ via MQTT.

Tasks:

- Connect ESP32 to Wi-Fi.
- Read data from IR sensor or analog input (potentiometer).
- Format data as JSON and publish to HiveMQ.
- Use MQTT Explorer or Node-RED to view values

Deliverables:

- Working code
 - Brief report
 - Max 3-min demo video
 - Circuit design
 - Wokwi simulation
-

✓ Task 4: Node-RED + IR Sensor + Servo (MQTT)

🧠 Objective:

Control a servo motor based on IR sensor data via Node-RED logic.

🔧 Tasks:

- Connect IR sensor to ESP32.
- Publish readings via MQTT (HiveMQ).
- Use Node-RED to:
 - Filter/process sensor values
 - Send commands back to ESP32 to control the servo

📦 Deliverables:

- Working code
 - Brief report
 - Max 3-min demo video
 - Circuit design
 - Wokwi simulation
-

✓ Task 5: Advanced MQTT + LCD Display

🧠 Objective:

Build a secure, resilient, scalable MQTT system with sensor data displayed on an LCD screen.

🔧 Part 1: MQTT Transmission

- Send real sensor data (IR) to HiveMQ.
- Format data in JSON.

Part 2: Display System

- Show MQTT data on LCD.
- Add local storage for disconnection handling in excel sheet.
- (Bonus) Add buttons/touch for interaction.

Deliverables:

- Working code
 - Brief report
 - Max 3-min demo video
 - Circuit design
 - Wokwi simulation
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Bonus Task 6: Dioty IoT App

Objective:

Create an IoT dashboard using the **Dioty** platform to monitor/control ESP32 devices.

Tasks:

- Set up Dioty account
- Send sensor data to Dioty using APIs (Open source)
- Build simple dashboard to:
 - Show live readings
 - Control a virtual or physical device (e.g., LED)

Deliverables:

- Working code
 - Brief report
 - Max 3-min demo video
 - Circuit design
 - Wokwi simulation
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Double Bonus: ESP32 + MQTT + Node-RED + InfluxDB + Grafana

Objective:

Create an end-to-end IoT pipeline:

ESP32 → MQTT (HiveMQ) → Node-RED → InfluxDB → Grafana

Tasks:

- ESP32 publishes sensor data to HiveMQ (IR Sensor)
- Node-RED subscribes and pushes to InfluxDB
- Grafana visualizes data as charts/graphs

Deliverables:

- Working code
- Brief report
- Max 3-min demo video
- Circuit design
- Wokwi simulation